Batterjee Medical College
For Science and Technology

BMC

Psychiatry · Clinical Lecture

Psychotic Disorders & Schizophrenia

What psychosis is, what causes it, how the psychotic disorders are separated from one another — and the principles of treating them.

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Learning Objectives (1 of 2)

Mapped to the BMC Knowledge / Skill / Value framework. Prerequisite: psychiatric history taking and the mental status examination.

Knowledge & understanding

- Define psychosis and list its signs and symptoms
- Classify delusions and hallucinations
- Name the psychotic disorders and the duration that separates them

Skills

- Elicit delusions, hallucinations and passivity phenomena in a history
- Distinguish psychosis from delirium at the bedside
- Build a differential for a first psychotic episode

Values & attitudes

- Treat psychosis as a medical illness, not a moral or spiritual failing
- Separate culturally sanctioned belief from delusion
- Involve the patient and the family in every decision

Learning Objectives (2 of 2)

Knowledge & understanding

- State the DSM-5-TR criteria for schizophrenia
- Outline the main aetiological hypotheses
- Describe the course, prognosis and principles of treatment

Skills

- Identify the medical and drug causes that must be excluded
- Recognise the extrapyramidal and metabolic harms of antipsychotics
- Outline treatment across the acute, stabilisation and maintenance phases

Values & attitudes

- Assess physical health as carefully as mental health
- Use non-stigmatising language in the record and with families
- Recognise and address the stigma these patients carry

Why this matters to you: whichever specialty you enter, you will prescribe drugs that cause psychosis, and you will be asked to see confused patients on your own ward.

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PART 1

What psychosis is

The definition · delusions · hallucinations · disorganised thought and behaviour · catatonia · negative symptoms.

A 21-year-old university student, brought by his father

Fahad is a second-year engineering student. Over the past eight months his family have watched him withdraw — he stopped meeting friends, his grades collapsed, and he now spends most of the day in his room.

For the last two months he has been convinced that a neighbour has placed a device in the flat to monitor him. He has covered the ceiling vent with tape. He hears two male voices commenting on what he is doing, and sometimes ordering him to stay silent.

He is not confused. He knows the date and where he is. He has never used substances, takes no medication, and has no medical history. His father says an uncle "had the same illness" in his twenties.

Keep Fahad in mind for the next two hours. By the end you should be able to name every symptom he has, justify the diagnosis, exclude the alternatives, and outline what you would do for him at each phase of treatment.

> **Notice what is NOT here: no clouding of consciousness, no fluctuation, no substance, no medical illness — and eight months of declining function.**

Psychosis and psychotic disorder

Two terms students routinely confuse. One is a symptom; the other is a group of diagnoses in which that symptom dominates.

Psychosis — the symptom

- A gross impairment in REALITY TESTING
- The inability to distinguish what is real from what is not
- A symptom, not a disease — like fever or breathlessness
- Caused by anything that disturbs brain function
- A patient with psychosis is described as "psychotic"

Psychotic disorders — the diagnoses

- A collection of disorders in which psychosis DOMINATES the symptom complex
- Brief psychotic disorder · schizophreniform · schizophrenia
- Schizoaffective disorder · delusional disorder
- Psychosis also appears in mood disorders, delirium, dementia and substance use
- It appears in the most severe forms of psychiatric illness

The clinical consequence: "psychotic" on a referral tells you nothing about the diagnosis. It is the starting point of the work-up, not the end of it.

When psychosis-like experience is normal

Three situations in which perceptual experiences occur in people who are not ill — know them, or you will over-diagnose.

Hypnagogic & hypnopompic

Vivid perceptions on FALLING ASLEEP (hypnagogic) or WAKING (hypnopompic). Common, brief, and confined to the sleep–wake transition. Also seen in narcolepsy.

Bereavement

Hearing or briefly seeing the deceased in the weeks after a death is a recognised, culturally normal grief phenomenon. The person knows it is not real, and function is otherwise preserved.

Severe sleep deprivation

Sustained sleep loss produces misperceptions and, at the extreme, frank hallucinations in healthy people. Reverses fully with sleep — a common finding in residents on long shifts.

The discriminators: insight is preserved, the experience is brief and situation-bound, and there is no decline in functioning.

The six domains of psychosis

Every psychotic presentation is described using these six. Learn them as a checklist — they reappear as Criterion A for schizophrenia.

1 · Delusions

Fixed, false beliefs that are not shakeable by evidence and are not shared by the person's culture or religion.

2 · Hallucinations

Perceptions in any of the five senses occurring WITHOUT an external stimulus. Auditory is by far the commonest in schizophrenia.

3 · Disorganised speech

Speech that reflects disorganised THOUGHT — derailment, tangentiality, incoherence. What you hear is a window onto thought form.

4 · Disorganised behaviour

Behaviour that is childlike, bizarre, unpredictable or aggressive, with loss of goal-directedness — including self-care.

5 · Catatonic behaviour

A marked reduction or peculiarity of motor activity: catalepsy, mutism, echolalia (repeating speech), echopraxia (repeating movements).

6 · Negative symptoms

Absence of what should be there — flat affect, poverty of speech, of motivation, of pleasure and of social contact.

Domains 1–3 are the POSITIVE symptoms and respond best to medication; domain 6 is the one that determines long-term disability and responds least.

Delusions — the definition first

Before naming the type, satisfy the definition. Most "delusions" reported by students are overvalued ideas or cultural beliefs.

A delusion is

- A FIXED, FALSE belief
- Held with absolute conviction despite contradictory evidence
- NOT accepted by the person's culture, religion or subculture
- Not amenable to reasoning or argument
- Often, though not always, acted upon

Bizarre versus non-bizarre

- BIZARRE — clearly implausible: "a chip has been implanted in my brain"
- NON-BIZARRE — possible but false: "my wife is having an affair"
- Non-bizarre delusions are the hallmark of DELUSIONAL DISORDER
- An OVERVALUED IDEA is unreasonable but held with some doubt — not a delusion
- An OBSESSION is resisted and recognised as one's own — not a delusion

The culture clause carries real weight in our practice — see the next slides. Source: DSM-5-TR, 2022.

Types of delusion — by theme

Type	The belief	Classically seen in
Persecutory (paranoid)	That one is being followed, watched, plotted against or harmed	The commonest type — schizophrenia, delusional disorder
Grandiose	That one is special, chosen, immensely wealthy or has special powers	Manic episode of bipolar disorder; schizophrenia
Nihilistic	That one is dead or dying, or that the world or one's organs no longer exist	Psychotic depression, especially in the elderly (Cotard)
Somatic	That a body part or function is missing or diseased — with no evidence: "my kidney was removed"	Psychotic depression; schizophrenia; delusional disorder, somatic type
Religious	A belief with religious content — typically a special relationship with God, usually with grandiosity	Schizophrenia; mania. Judge against the person's own community

A scar-free abdomen and a normal ultrasound do not shift the somatic delusion — that unshakeability is what makes it a delusion rather than health anxiety.

More types — and the passivity phenomena

Type	The belief	Note
Erotomanic	That a person of higher status or a public figure is in love with the patient	De Clérambault syndrome; may lead to stalking
Jealous	That the spouse or partner is unfaithful, on no evidence	Othello syndrome; associated with alcohol misuse and with violence risk
Of reference	That external cues are uniquely aimed at the patient: "the presenter is talking to me"	Very common; often the earliest symptom families notice
Thought broadcasting	That one's thoughts can be heard by other people	A passivity phenomenon — Schneiderian first-rank symptom
Thought insertion / withdrawal	That thoughts are being put into, or removed from, one's mind by an outside force	Passivity — the boundary of the self is breached
Delusion of control	That one's actions, feelings or impulses are being directed by an external agent	Also called made acts / made feelings

The last three are PASSIVITY phenomena — the patient experiences their own mental life as coming from outside. Ask about them explicitly; patients rarely volunteer them.

Belief, culture and religion

A belief is only a delusion if it falls outside what the patient's own community accepts. In our setting this judgement is made almost daily.

Not a delusion

- Belief in the evil eye, jinn, sihr (magic) or ruqyah where the community shares it
- Seeking a religious healer before a doctor — a normal help-seeking pathway here
- Intense religious practice within the person's own tradition
- Attributing an illness to a spiritual cause while accepting treatment

Points toward a delusion

- A belief the patient's OWN family and imam say is not normal for them
- Content that is idiosyncratic and personal rather than shared
- A special personal role: "I have been chosen to judge people"
- Accompanied by hallucinations, disorganisation and functional decline
- Held with a conviction that isolates the patient from their community

The practical test: ask the family. "Is this normal for him, or for people you know?" The community, not the doctor, defines the baseline.

Is this a delusion?

A 34-year-old man is brought by his brother. He says the family has been affected by the evil eye and has asked for ruqyah. He works normally, prays with the family and sleeps well.

His brother says the whole extended family holds the same belief. There are no hallucinations and no decline in function. The referral letter reads: "delusional beliefs — assess for antipsychotics."

What is the correct assessment?

A A delusion — start an antipsychotic

B A culturally shared belief — not a delusion

C An overvalued idea — start CBT

D A somatic delusion — investigate

Commit to an answer before the next slide.

Is this a delusion?

ANSWER

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C An overvalued idea — start CBT

D A somatic delusion — investigate

B — the belief is shared by his community, and nothing else is wrong.

A delusion must be false, fixed AND outside the person's cultural or religious frame. Here the belief is shared by the extended family, there are no hallucinations, no disorganisation and no functional decline — the diagnostic criteria are simply not met. Treating him would expose him to metabolic and extrapyramidal risk for no benefit, and would lose the family's trust. What would change the answer: an idiosyncratic version the family themselves find strange, added hallucinations, or a decline in work and self-care.

Hallucination, illusion, pseudohallucination

Three terms, one distinction: is there a stimulus, and is the experience located in outside space or inside the head?

Hallucination

- A perception with NO external stimulus
- Experienced as real and located in external space
- Occurs in any of the five sensory modalities
- Insight is typically absent

Illusion

- A MISPERCEPTION of a real external stimulus
- The stimulus exists — it is misread
- A dressing-gown on a door seen as a person
- Common in delirium, poor light and high anxiety

Pseudohallucination

- Perceived in INNER subjective space — "inside my head"
- Insight is usually retained
- Seen in dissociation, personality disorder, grief
- Not by itself evidence of a psychotic disorder

A true hallucination is located out there in the world. That single question — "where does the voice come from?" — separates the first column from the third.

Hallucinations by modality

Modality	What the patient describes	What it should make you think
Auditory	Voices — commonly two or more talking about the patient, or commanding them	The commonest in schizophrenia. Command hallucinations = ask about compliance with them
Visual	Seeing people, animals or figures that are not there	Less common in schizophrenia — think DELIRIUM, dementia (Lewy body), substances, epilepsy
Olfactory	An unpleasant smell, often burning, preceding other symptoms	Classically an aura of TEMPORAL LOBE EPILEPSY — investigate
Tactile	Insects crawling on or under the skin (formication)	Stimulant intoxication (cocaine, amphetamine) and alcohol withdrawal
Gustatory	A persistent strange or metallic taste	Rare — epilepsy, and some drug effects

The rule that saves lives: a NON-AUDITORY hallucination in a first presentation is organic until proven otherwise. Visual and olfactory hallucinations demand a medical work-up before a psychiatric label.

Thought form — the ladder of severity

Disorganised speech is the only window we have onto disorganised THOUGHT — the terminology describes the form, not the content.



Ask yourself one question as the patient speaks: can I still follow how they got from one idea to the next? That is the whole assessment of thought form.

Disorganised behaviour and catatonia

The motor domain of psychosis. The second column is the one that kills if it is missed.

Disorganised behaviour

- Childlike silliness; bizarre or compulsive acts
- Unpredictable or aggressive behaviour
- Loss of goal-directed behaviour — including self-care and hygiene
- Inappropriate dress and collapse of daily routine
- Judge it against what is normal for that person, not for you

Catatonia

- CATALEPSY / waxy flexibility — limbs held in imposed postures
- MUTISM; negativism; posturing; purposeless excitement
- ECHOLALIA (repeating your words); ECHOPRAXIA (copying your movements)
- A MEDICAL EMERGENCY — and commonest in MOOD disorders, not schizophrenia
- No longer a subtype of schizophrenia; it crosses diagnoses

Catatonia responds to a benzodiazepine (lorazepam) challenge and to ECT. Its dangerous form — malignant catatonia with fever and autonomic instability — is clinically indistinguishable from neuroleptic malignant syndrome.

Negative symptoms — the five A's

Positive symptoms are what is added; negative symptoms are what has been lost. The second group determines long-term disability.

Affective flattening

Blunted or flat affect. Blunted = reduced range; flat = virtually absent. The face stops carrying emotion, and the voice loses its melody.

Alogia

Poverty of speech — brief, empty replies; or poverty of the CONTENT of speech, where much is said but little information is conveyed.

Avolition & abulia

Poverty of movement and of motivation. Goal-directed activity collapses: work, study and self-care stop. Routinely mistaken for laziness.

Anhedonia

Poverty of joy — the loss of pleasure in things previously enjoyed. Shared with depression, which is why the two are confused.

Asociality

Poverty of socialisation — withdrawal from friends and family, and loss of interest in social contact rather than avoidance through anxiety.

Why they matter

They respond LEAST to antipsychotics, cause most of the functional impairment, and are the strongest predictor of poor long-term outcome.

Differential: depression, the extrapyramidal side effects of antipsychotics, and institutionalisation all mimic negative symptoms. Exclude those before accepting them as primary.

Schneider's first-rank symptoms

Once thought to be specific to schizophrenia. They are not — but they remain heavily examined and are still clinically useful prompts.

The first-rank symptoms

- Thought INSERTION — thoughts placed into the mind from outside
- Thought WITHDRAWAL — thoughts removed from the mind
- Thought BROADCASTING — thoughts audible or known to others
- Third-person auditory hallucinations — voices discussing the patient
- Running commentary on the patient's actions; made acts, feelings and impulses

Where they now stand

- Present in roughly 3 of 4 patients with schizophrenia
- But also in mania, psychotic depression and organic psychoses
- DSM-5 REMOVED their special diagnostic weight — they no longer count double
- Still valuable as a structured checklist when you take a history
- Their presence should never end the search for an organic cause

DSM-IV allowed a single bizarre delusion or a running commentary to satisfy Criterion A alone. DSM-5-TR abolished that — two symptoms are now always required. Source: DSM-5-TR, 2022.

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PART 2

Before you call it schizophrenia

Delirium · medical causes · substances and prescribed drugs · the work-up that must come first.

Day 3 after surgery

A 72-year-old man is three days post hemiarthroplasty. The night nurse calls: he is pulling at his cannula, says there are children in the room, and does not know where he is.

He was fully oriented this morning and briefly drowsy last night. He takes tramadol and oxybutynin; temperature 37.9 °C. The surgical team ask you to "refer to psychiatry for acute psychosis".

What is the most likely explanation?

A Late-onset schizophrenia

B Delirium — find the cause

C Brief psychotic disorder

D Delusional disorder, persecutory type

Commit to an answer before the next slide.

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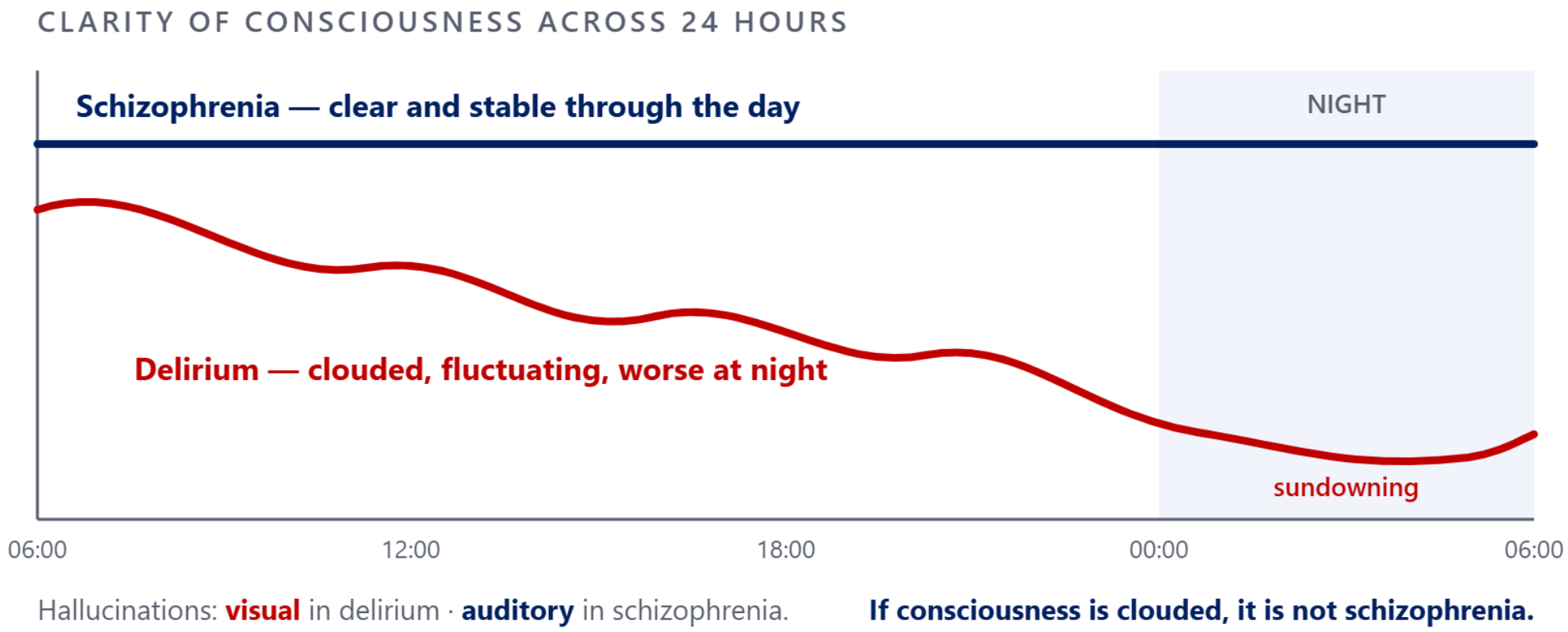
C Brief psychotic disorder

D Delusional disorder, persecutory type

B — this is delirium until proved otherwise, and it is a medical emergency.

Five features make the diagnosis: ACUTE onset over hours, FLUCTUATION across the day, IMPAIRED consciousness and attention, DISORIENTATION, and VISUAL hallucinations. None of these belong to schizophrenia, in which consciousness and orientation are preserved. He also has two anticholinergic-burden drugs and a fever. The correct response is not a psychiatric referral: examine him, screen for infection and hypoxia, check glucose, electrolytes, calcium and renal function, review every drug, and treat the cause. Delirium in an elderly surgical patient carries a mortality comparable to myocardial infarction.

Delirium versus a primary psychosis

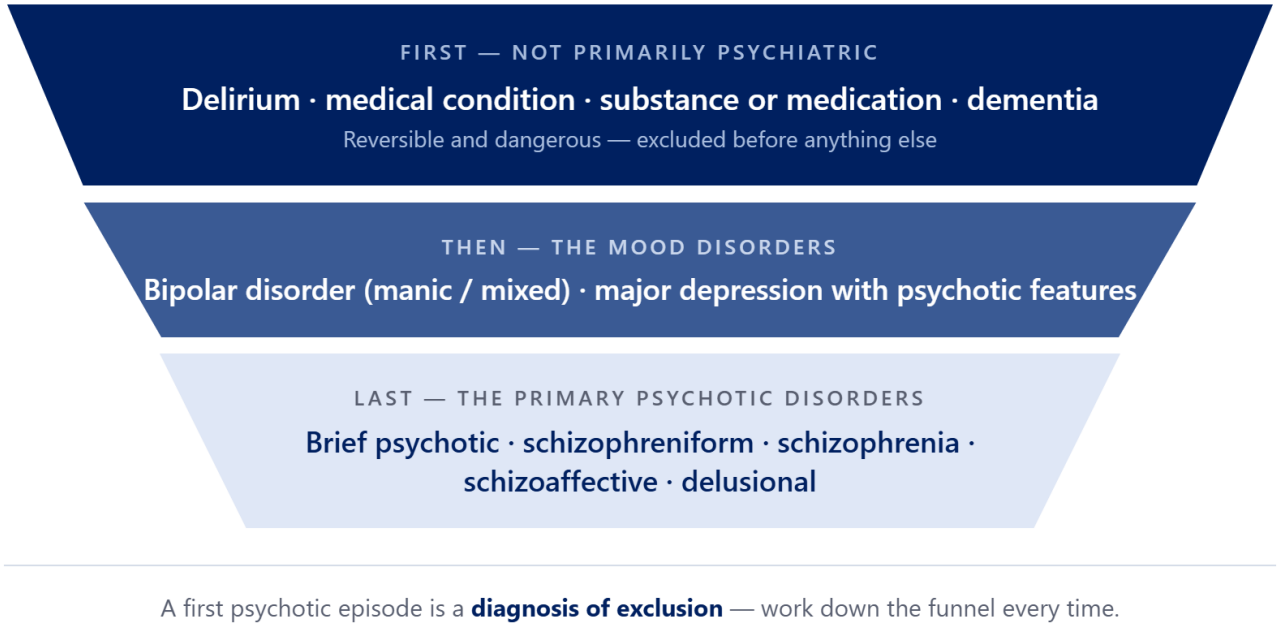


If consciousness is clouded, it is not schizophrenia. Two clear rules for the ward: any new psychosis after age 40 is organic until proven otherwise, and any fluctuating presentation is delirium until proven otherwise.

The differential diagnosis of psychosis

THINK IN THIS ORDER

Work down the funnel every time. The order is deliberate — the reversible and dangerous causes come first, and the label you were reaching for comes last.



A first psychotic episode is a **diagnosis of exclusion** — work down the funnel every time.

A first psychotic episode is a diagnosis of exclusion. "Schizophrenia" is applied only after the tiers above it have been actively excluded, not assumed away.

Psychosis due to a medical condition

Group	Examples
CNS disease	Cerebrovascular disease · multiple sclerosis · brain tumour · Alzheimer and Parkinson disease · Huntington disease · epilepsy (especially temporal lobe) · encephalitis · neurosyphilis · prion disease · neurosarcoidosis · HIV/AIDS
Endocrine	Hyper- and hypothyroidism · Cushing and Addison disease · hyper- and hypocalcaemia · hypopituitarism
Nutritional	Vitamin B12 deficiency · folate deficiency · niacin deficiency (pellagra) · thiamine deficiency
Autoimmune	Systemic lupus erythematosus · temporal arteritis · ANTI-NMDA RECEPTOR ENCEPHALITIS — psychosis, seizures, movement disorder and autonomic instability in a young patient
Metabolic / other	Porphyria · hepatic and uraemic encephalopathy · hypoglycaemia · hypoxia · Wilson disease

Anti-NMDA receptor encephalitis deserves its own mention: it often presents first to psychiatry, and it is treatable. Suspect it when psychosis in a young patient is accompanied by seizures, dyskinesias or autonomic instability.

Substance- and drug-induced psychosis

Two groups: what the patient takes, and what the doctor gives. The second is the one students never consider.

Substances of misuse

- AMPHETAMINE-type stimulants (including Captagon) and cocaine — during intoxication
- Cannabis — dose-dependent, and a risk factor for later schizophrenia
- Hallucinogens (LSD, MDMA) and phencyclidine (PCP)
- Alcohol, benzodiazepine and barbiturate WITHDRAWAL
- Inhalants; toxins — carbon monoxide, organophosphates

Prescribed medication

- CORTICOSTEROIDS — dose-related; a common cause on medical wards
- Anti-parkinsonian dopaminergic drugs (levodopa, dopamine agonists)
- ANTICHOLINERGICS and antihistamines — especially in the elderly
- Anticonvulsants, antimicrobials (isoniazid, fluoroquinolones, mefloquine)
- Stimulants (methylphenidate), interferon, chemotherapy agents, digoxin

Substance-induced psychosis typically resolves within days to weeks of abstinence. If it persists well beyond intoxication and withdrawal, reconsider the diagnosis — stimulants can unmask a primary psychotic illness.

Red flags: this is not schizophrenia

Any one of these should stop you from making a primary psychiatric diagnosis until the work-up is complete.

Age at onset over 40

First-episode psychosis after 40, and especially after 60, is organic until proven otherwise. Schizophrenia almost always declares itself in the teens to thirties.

Non-auditory hallucinations

Visual, olfactory, tactile or gustatory hallucinations point to delirium, epilepsy, a lesion or a substance — not to schizophrenia.

Clouded or fluctuating consciousness

Drowsiness, disorientation, an inattentive patient, or a picture that is worse at night. This is delirium.

Abnormal neurological signs

Focal deficits, seizures, abnormal movements, incontinence or a new headache demand imaging before a psychiatric label.

Fever or autonomic instability

Consider encephalitis, sepsis, malignant catatonia and neuroleptic malignant syndrome. All are emergencies.

Very rapid onset in hours

Psychosis that appears over hours in a previously well person suggests a drug, a toxin, or an acute medical event.

None of these excludes a psychiatric illness — but each one obliges you to look for something else first.

The first-episode work-up

Investigation	Looking for
History, collateral and full physical examination	The single highest-yield step — timeline, drugs, medical history, family account. Never diagnose a first episode without collateral
CBC, renal and liver profile, electrolytes, calcium, glucose	Infection, uraemia, hepatic encephalopathy, hyper/hypocalcaemia, hypoglycaemia
TSH, vitamin B12, folate	Thyroid disease and nutritional deficiency — common, cheap and reversible
Urine drug screen	Amphetamines, cannabis, cocaine and opioids — required in every first episode
HIV and syphilis serology; ESR/CRP, ANA if indicated	Infective and autoimmune causes, including neurosyphilis and SLE
CT or MRI brain; EEG if indicated	Tumour, stroke, demyelination; EEG for suspected temporal lobe epilepsy or encephalitis
Baseline before antipsychotics: weight, BMI, waist, BP, lipids, HbA1c, ECG	You cannot detect metabolic or QT harm later without a baseline recorded now

The last row is the one most often skipped, and the one you will regret. Record the baseline before the first dose.

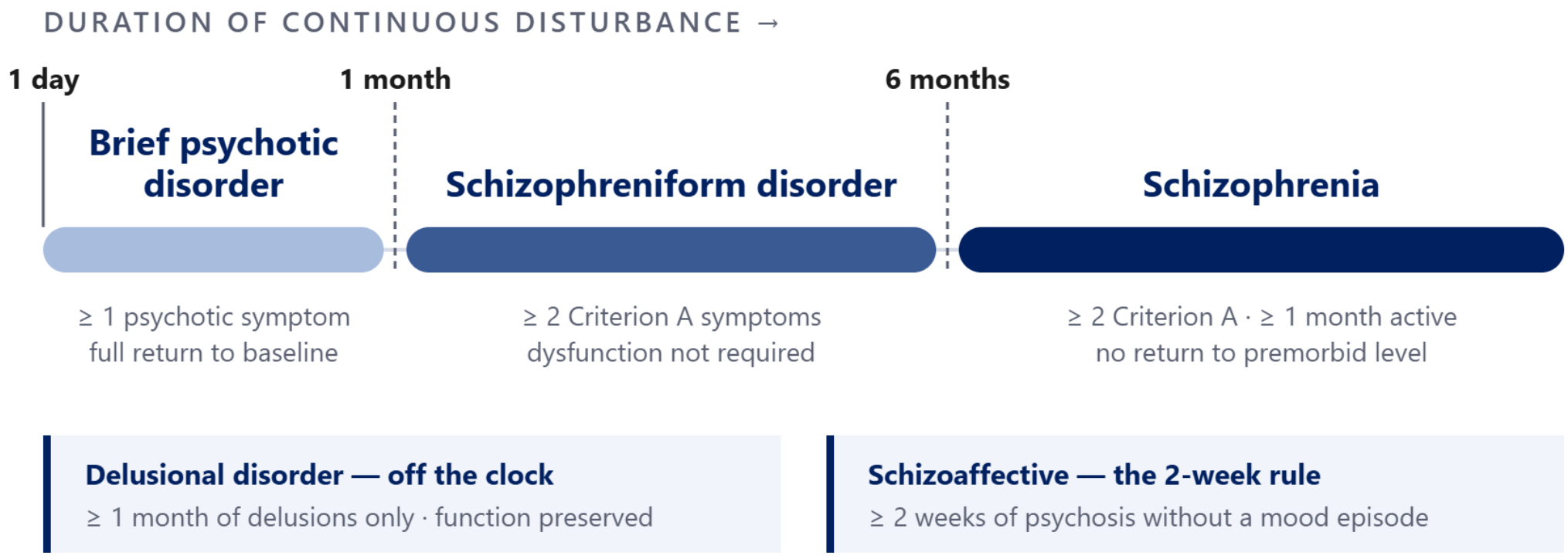
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PART 3

The psychotic disorders

Brief psychotic disorder · schizophreniform · schizoaffective · delusional disorder — and the clock that separates them.

The psychotic disorders — the clock



Delusional disorder — off the clock

≥ 1 month of delusions only · function preserved

Schizoaffective — the 2-week rule

≥ 2 weeks of psychosis without a mood episode

Two diagnoses sit off this axis. DELUSIONAL DISORDER: ≥ 1 month of delusions with no prominent hallucinations and function preserved outside the belief. SCHIZOAFFECTIVE: ≥ 2 weeks of psychosis WITHOUT a mood episode, with mood symptoms for most of the illness. Treatment does not wait for the label — all are treated as a first psychotic episode from day one.

Brief psychotic and schizophreniform disorders

Brief psychotic disorder

- Positive psychotic symptoms lasting 1 day to < 1 month
- At least ONE of: delusions, hallucinations, disorganised speech, disorganised or catatonic behaviour
- FULL return to the premorbid level of functioning
- Often follows a marked stressor; a postpartum-onset specifier exists
- Risk of recurrence and of later reclassification is real — follow these patients up

Schizophreniform disorder

- TWO or more Criterion A symptoms lasting 1 to < 6 months
- Identical to schizophrenia except: duration < 6 months, and social or occupational dysfunction is NOT required
- Full return to premorbid functioning if it resolves
- Lifetime prevalence about 0.2%; men and women equally affected
- If symptoms pass 6 months, the diagnosis becomes schizophrenia

Treatment does not wait for the label. Both are treated as a first psychotic episode from day one; the diagnosis is revised as the clock runs.

Schizoaffective disorder

Defined by a two-week rule. Getting this wrong in either direction changes treatment for years.

The diagnosis requires

- A major mood episode — depressive or manic — CONCURRENT with Criterion A symptoms of schizophrenia
- Delusions or hallucinations for ≥ 2 WEEKS in the ABSENCE of a prominent mood episode
- Mood symptoms present for the MAJORITY of the total illness duration
- Not attributable to a substance or another medical condition
- Lifetime prevalence about 0.3% (DSM-5-TR); more common in women

How to tell it apart

- MOOD DISORDER WITH PSYCHOTIC FEATURES — psychosis occurs ONLY during the mood episode
- SCHIZOPHRENIA — mood symptoms are brief or absent relative to the psychosis
- SCHIZOAFFECTIVE — psychosis persists ≥ 2 weeks after the mood episode settles
- The whole distinction rests on a careful longitudinal timeline
- Draw the timeline with the family; it cannot be made from a single interview

Older texts quote a lifetime prevalence of 0.5–0.8%; DSM-5-TR gives approximately 0.3%. Both figures are estimates from different case definitions.

Delusional disorder

The features

- ONE or more delusions lasting ≥ 1 month
- Typically NON-BIZARRE — the belief is possible, just false
- No prominent hallucinations; Criterion A of schizophrenia is not met
- The delusion is ENCAPSULATED — the rest of mental life is intact
- Behaviour is not obviously odd; functioning outside the delusion is preserved

Subtypes and pitfalls

- Persecutory · jealous · erotomanic · somatic · grandiose · mixed
- Patients rarely present themselves — the family or the police bring them
- Insight is absent, so engagement is the hardest part of management
- The jealous subtype carries a documented risk of violence to the partner
- The somatic subtype often reaches dermatology or surgery long before psychiatry

These patients function, work and hold conversations normally — which is exactly why the diagnosis is missed for years, and why confrontation destroys the therapeutic relationship.

Which diagnosis?

A 29-year-old woman has had persecutory delusions and voices for five months. During the first three months she was also profoundly depressed — early waking, no appetite, hopeless, unable to work.

The depression has now settled with treatment. For the last six weeks her mood has been reasonable, but the voices and the belief that she is being followed continue unchanged.

What is the most likely diagnosis?

A Depression with psychotic features

B Schizophrenia

C Schizoaffective disorder, depressive type

D Brief psychotic disorder

Commit to an answer before the next slide.

Which diagnosis?

ANSWER

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What is the most likely diagnosis?

A Depression with psychotic features

B Schizophrenia

C **Schizoaffective disorder, depressive type**

D Brief psychotic disorder

C — psychosis has persisted well beyond two weeks without a prominent mood episode.

Option A fails because in mood disorder with psychotic features the psychosis occurs ONLY within the mood episode; here it continued for six weeks after the depression settled. Option B fails because a full major depressive episode was present for the majority of the illness — that is not typical of schizophrenia. Option D fails on duration; brief psychotic disorder must resolve within a month. The two-week rule is what decides this case, and it is why the longitudinal timeline — not the cross-sectional picture — makes the diagnosis.

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PAUSE

Break — 10 minutes

After the break: schizophrenia itself — epidemiology, aetiology, the DSM-5-TR criteria, course, prognosis and treatment.

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PART 4

Schizophrenia

Who gets it, and why — epidemiology, the stress–diathesis model, genetics, dopamine and the other hypotheses.

Epidemiology of schizophrenia

	Figure	Comment
Lifetime prevalence	≈ 1%	The classically taught figure. DSM-5-TR gives a narrower lifetime estimate of about 0.3–0.7%
Age at onset — men	≈ 21 years	Late teens to mid-twenties; onset before puberty or after 45 is rare
Age at onset — women	≈ 27 years	A second, smaller peak occurs around the menopause
Sex ratio	M ≥ F	Slight male excess in most studies; some report equal rates with a later female onset
Male course	More severe	Earlier onset, more prominent NEGATIVE symptoms, poorer treatment response, poorer prognosis
Female course	Less severe	Later onset, more prominent POSITIVE symptoms, better treatment response, better prognosis
Life expectancy	15–20 years shorter	Driven mainly by cardiovascular and metabolic disease and smoking — not by suicide alone

The 1% figure is easy to remember and roughly right; the narrower DSM-5-TR range is the more defensible number. Both are given here rather than collapsing them into one.

The stress–diathesis model

No single factor causes schizophrenia. A vulnerability (diathesis) is necessary; a stressor is what converts vulnerability into illness.

Diathesis — the vulnerability

- Genetic loading — the largest single contribution
- Dopaminergic, serotonergic and glutamatergic dysfunction
- Neurodevelopmental insult — obstetric complications, maternal infection, winter birth
- Structural and functional brain changes present before onset
- Present long before any symptom appears

Stress — the trigger

- Substance use, especially cannabis and stimulants in adolescence
- Bereavement or the loss of a significant relationship
- Migration, urban upbringing and social adversity
- Major environmental or role change — leaving home, starting university
- High expressed emotion at home — a potent trigger for RELAPSE

Neither column alone is sufficient. This model is also how you explain the illness to a family: the vulnerability was there; the stressor was the last push, not the cause.

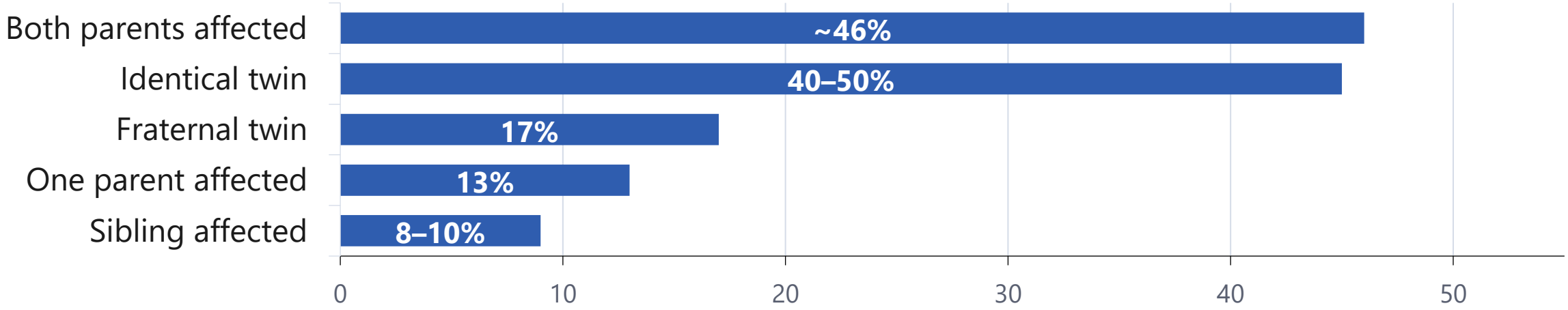
Genetic risk rises with closeness

50–70%
of the risk of developing schizophrenia is attributed to genetic factors

40–50%
concordance in identical twins — so genes are not the whole story

~1%
baseline lifetime risk in the general population

Lifetime risk of schizophrenia by relationship to an affected person (%) — against a general-population baseline of about 1%



Where a source gives a range, the bar is drawn to the middle of it and the LABEL carries the full published range. The decisive observation is the identical-twin figure: genetically identical people are concordant less than half the time, so environment must matter. Source: Kaplan & Sadock's Synopsis of Psychiatry.

The dopamine hypothesis

Excess dopamine in the mesolimbic pathway produces the positive symptoms. Blocking D2 receptors treats them — and produces the side effects, in the other three pathways.

Mesolimbic — the target

- Hyperdopaminergic state → POSITIVE symptoms
- D2 blockade here is the therapeutic effect

Mesocortical — under-active

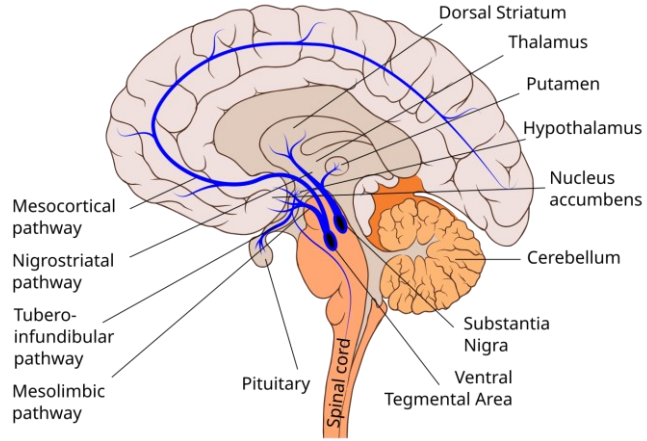
- Low dopamine → negative and cognitive symptoms
- Blocking it further can WORSEN these

Nigrostriatal — motor

- D2 blockade → extrapyramidal side effects and tardive dyskinesia

Tuberoinfundibular — endocrine

- D2 blockade → raised prolactin: galactorrhoea, amenorrhoea, sexual dysfunction



Dopaminergic pathways. Wikimedia Commons — User:Slashme, after Patrick J. Lynch. CC BY-SA 4.0.

The pharmacological proof: cocaine and amphetamine raise dopamine and EXACERBATE psychosis; first- and second-generation antipsychotics block D2 and REDUCE it.

Beyond dopamine

Dopamine explains the positive symptoms and the drug effects, but not the negative symptoms, the cognitive deficits or the developmental timing.

Serotonin & glutamate

- 5-HT2A antagonism underlies the second-generation antipsychotics
- NMDA receptor HYPOFUNCTION is the leading alternative model
- Ketamine and PCP block NMDA receptors and reproduce positive, negative AND cognitive symptoms in healthy volunteers
- The best pharmacological model of the negative symptoms we have

Neurodevelopmental

- Obstetric complications, maternal influenza and famine raise risk
- Excess winter and early-spring births
- Premorbid cognitive and motor abnormalities appear in childhood
- The lesion is laid down early; symptoms emerge in adolescence

The neurodevelopmental account explains the timing that dopamine cannot: why an illness with prenatal risk factors declares itself only in late adolescence.

Brain changes and inflammation

What imaging and immunology add — and, just as importantly, what they do not allow you to do.

Structural & functional changes

- ENLARGED LATERAL VENTRICLES and reduced total brain volume
- Accelerated loss of grey matter through adolescence
- Reduced hippocampal, amygdala and temporal lobe volume
- HYPOFRONTALITY — reduced frontal activation on functional imaging
- Present at first episode, before any antipsychotic exposure

Autoimmune & inflammatory

- Raised inflammatory markers in a subgroup of patients
- ANTI-NMDA RECEPTOR ENCEPHALITIS is the proof of principle — an antibody-mediated psychosis that responds to immunotherapy
- Suspect it with seizures, dyskinesias or autonomic instability in a young patient
- Not a routine screen — but a diagnosis not to miss

None of these findings is diagnostic. Imaging in schizophrenia is done to EXCLUDE other causes, not to confirm the diagnosis.

Mechanism to side effect

A 24-year-old woman started haloperidol three weeks ago for a first psychotic episode. The voices have almost gone.

She now reports that her periods have stopped and that a milky discharge comes from both breasts. A pregnancy test is negative and her prolactin is markedly raised.

Which dopamine pathway explains this?

A Mesolimbic

B Mesocortical

C Nigrostriatal

D Tuberoinfundibular

Commit to an answer before the next slide.

Mechanism to side effect

ANSWER

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She now reports that her periods have stopped and that a milky discharge comes from both breasts. A pregnancy test is negative and her prolactin is markedly raised.

Which dopamine pathway explains this?

A Mesolimbic

B Mesocortical

C Nigrostriatal

D **Tuberoinfundibular**

D — tuberoinfundibular. Dopamine normally INHIBITS prolactin release.

Dopamine travelling from the hypothalamus to the anterior pituitary tonically suppresses prolactin. Block those D2 receptors and prolactin rises, producing amenorrhoea, galactorrhoea, sexual dysfunction and — over years — reduced bone mineral density. Note what else the same case tells you: the improvement in her voices is the MESOLIMBIC effect of the same drug. Management options are to reduce the dose, or switch to a prolactin-sparing agent such as aripiprazole. Do not simply stop treatment because of this side effect — but do not ignore it either; it is a common and unspoken reason patients stop taking their medication.

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PART 5

Diagnosis

The DSM-5-TR criteria A to F · positive, negative and cognitive symptoms · the historical subtypes.

DSM-5-TR criteria for schizophrenia

ALL SIX MUST BE SATISFIED

A → F

Six criteria. A is the symptom criterion, B the functional one, C the clock, and D–F are exclusions. A diagnosis that skips the exclusions is not a diagnosis.

Source: DSM-5-TR, 2022.

A Characteristic symptoms

≥ 2 of: delusions · hallucinations · disorganised speech · grossly disorganised or catatonic behaviour · negative symptoms — for a significant part of 1 month. AT LEAST ONE must be from the first three.

B Functional decline

Functioning in work, relationships or self-care markedly below the level reached before onset — or, with childhood onset, expected levels never reached.

C Duration

Continuous disturbance for AT LEAST 6 MONTHS, including ≥ 1 month of Criterion A. The remainder may be prodromal or residual.

D Mood-disorder exclusion

Schizoaffective disorder and mood disorder with psychosis excluded — no major mood episode during the active phase, or only for a minority of the illness.

E Substance / medical exclusion

Not attributable to a substance or another medical condition — Part 2 of this lecture, written into the criteria.

F Relationship to autism

With a history of autism or a childhood communication disorder, diagnose schizophrenia only if prominent delusions or hallucinations are also present for ≥ 1 month.

Criterion A in practice

Two symptoms for one month, and at least one of them must come from the first three. Negative symptoms alone can never make the diagnosis.

The five Criterion A symptoms

- 1 · Delusions
- 2 · Hallucinations
- 3 · Disorganised speech
- 4 · Grossly disorganised or catatonic behaviour
- 5 · Negative symptoms

How the rule works

- At least TWO must be present for a significant part of one month
- At least one must be 1, 2 or 3 — the "core" psychotic symptoms
- So 4 + 5 together do NOT satisfy Criterion A
- Less than one month of symptoms treated successfully still counts
- DSM-IV's single-bizarre-delusion shortcut was ABOLISHED in DSM-5

Worked example: a patient with delusions and negative symptoms satisfies Criterion A. A patient with catatonia and negative symptoms alone does not.

Positive symptoms

What has been ADDED to normal experience. These are the symptoms that respond best to antipsychotic medication.

Delusions

Fixed, false beliefs outside the person's culture. Persecutory is commonest; reference, control and passivity phenomena are frequent.

Hallucinations

Predominantly AUDITORY — third-person voices discussing the patient, or a running commentary. Command hallucinations must be asked about directly.

Disorganised speech

Loosening of associations, tangentiality, incoherence, neologisms — the audible form of disorganised thought.

Disorganised behaviour

Unpredictable agitation, bizarre or childlike acts, inappropriate dress, collapse of goal-directed activity and self-care.

Catatonic symptoms

Stupor, mutism, negativism, posturing, waxy flexibility, echolalia, echopraxia — or, at the other extreme, purposeless excitement.

Why they matter clinically

They are the reason patients present, the symptoms families notice, and the ones most likely to improve within 2–4 weeks of starting treatment.

Negative symptoms

What has been LOST. Present in most patients, often preceding the first positive symptom by years.

Affective flattening

Blunted or flat affect — reduced facial expression, poor eye contact, monotonous voice, absent gesture.

Alogia

Poverty of speech and of the content of speech: brief, empty replies that convey little information.

Avolition & abulia

Poverty of movement and of motivation. Work, study and hygiene stop. This is the single largest driver of disability.

Anhedonia

Poverty of joy — loss of pleasure and of the anticipation of pleasure. Overlaps with depression, which must be excluded.

Asociality

Poverty of socialisation — reduced interest in and initiation of social contact, not primarily driven by anxiety.

Poor treatment response

Antipsychotics do little for these symptoms. First exclude the mimics: depression, drug-induced parkinsonism, and understimulation.

Primary negative symptoms are part of the illness; SECONDARY negative symptoms are caused by depression, medication or the environment — and those are treatable.

Cognitive symptoms

The least visible group, and the best predictor of whether a patient will return to work or study.

Attention

Inability to sustain attention, and heightened distractibility. The patient cannot hold a task or a conversation across interruptions.

Working memory & multitasking

Inability to hold and manipulate information — so multitasking, following instructions and planning a day all break down.

Executive function — SOAP

SEQUENCING · ORGANISATION · ABSTRACTING · PLANNING. Test with proverbs and similarities; concrete answers are typical.

Onset

Cognitive deficits are present in the premorbid phase, before the first psychotic symptom, and persist between episodes.

Prognostic weight

Cognitive impairment predicts functional outcome — employment, independent living — better than positive symptoms do.

Treatment

No agent reliably improves cognition. Cognitive remediation and supported employment are the interventions with evidence.

This is why a patient whose voices have completely resolved may still be unable to return to university. Explain that to the family early.

The historical subtypes

REMOVED IN DSM-5, STILL EXAMINED

Subtype	Defining feature
Catatonic	Presence of ≥ 1 catatonic symptom, regardless of what else is present
Disorganised (hebephrenic)	Disorganised speech and behaviour, flat or inappropriate affect — and NO catatonic symptoms
Paranoid	Delusions and hallucinations dominate, with no catatonic symptoms and no disorganised features. Best preserved functioning
Undifferentiated	Criterion A is met but the picture fits none of the above
Residual	No prominent positive symptoms; attenuated negative symptoms and odd beliefs persist

DSM-5 ABOLISHED these subtypes — they were unstable over time, overlapped heavily, and did not predict treatment response or outcome. They were replaced by dimensional severity ratings. Know them for exams and for reading older notes; do not use them to plan treatment. Source: DSM-5-TR, 2022.

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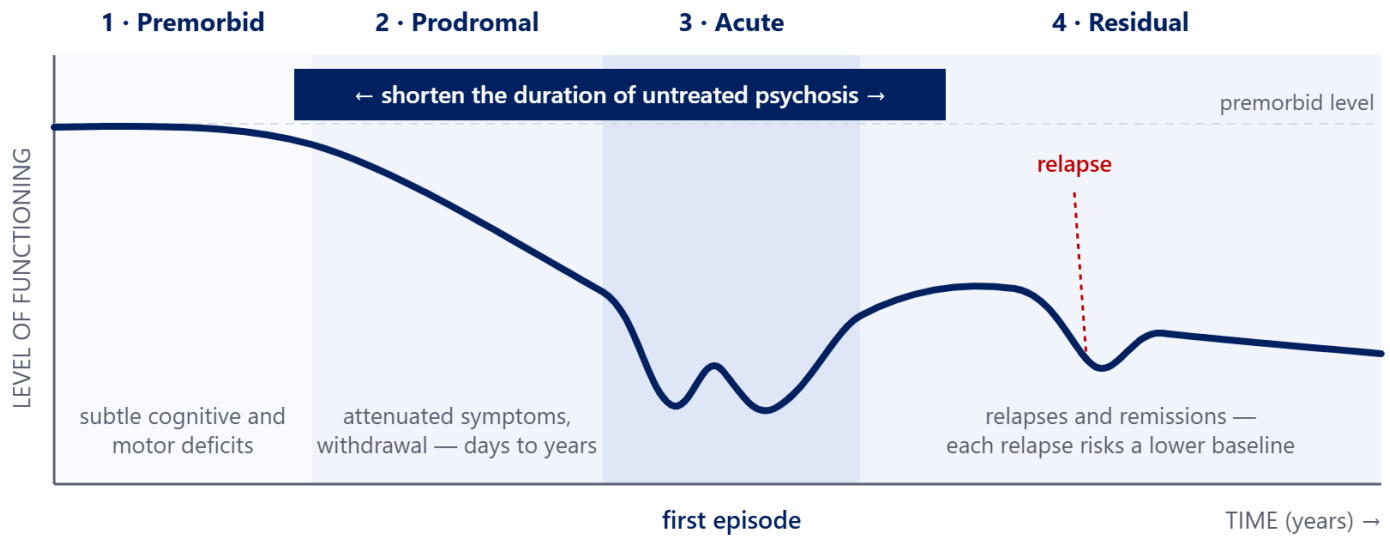
PART 6

Course & prognosis

The four phases · physical comorbidity and the mortality gap · what predicts a good or a poor outcome.

Phases 1 and 2 — before the diagnosis

Recognising the prodrome matters: shortening the duration of untreated psychosis is one of the few things that measurably improves outcome.



The prodrome — not the acute phase — is where the greatest gains are available.

Long duration of untreated psychosis predicts worse outcome — which is why the prodromal phase, not the acute one, is where the greatest gains are available.

Phases 3 and 4 — illness and aftermath

The acute phase is the one students see on the ward. The residual phase is the one the patient and the family live in.

3 · Acute (active)

- Criterion A symptoms — ≥ 2 psychotic symptoms
- Often follows a serious stressor or substance use
- Severe functional impairment
- Usually requires hospitalisation
- The phase in which patients most often first present

4 · Residual

- Follows resolution of the active-phase symptoms
- Attenuated positive and negative symptoms persist
- Depression and cognitive impairment are common
- Function remains below the premorbid level
- Relapse into a further acute phase is the main risk

Schizophrenia is a chronic illness of relapses and remissions. Each acute relapse risks leaving the patient at a lower functional baseline than before.

Physical health and the mortality gap

Schizophrenia is a chronic illness with relapses and remissions — and its greatest mortality burden is physical, not psychiatric.

Smoking

About three times more likely to smoke than the general population, and to smoke more heavily. The largest single contributor to premature death.

Metabolic disease

Higher rates of obesity, type 2 diabetes, dyslipidaemia and cardiovascular disease — from the illness, the lifestyle and the medication together.

Infection

Higher rates of HIV and other blood-borne infection, linked to risk behaviour and reduced access to preventive care.

Substance use

Alcohol and drug misuse are markedly more common, worsening adherence, relapse risk and physical health together.

Suicide

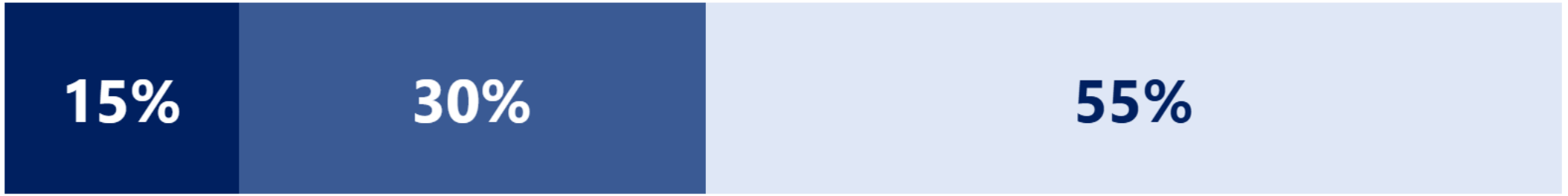
About 5% die by suicide (older texts quote 10–13%); risk is highest early in the illness, in young men, and in those with insight and depression.

The diagnostic trap

DIAGNOSTIC OVERSHADOWING — new physical symptoms in a psychotic patient get attributed to the psychiatric illness and are not investigated.

Suicide figures differ by source: Palmer et al. (2005) estimate ≈ 4.9% lifetime, against the 10–13% quoted in older textbooks. Both are shown rather than a single number.

Prognosis is not uniform



Good
substantial recovery of function

Intermediate
partial recovery, some independence

Poor
persistent symptoms and disability

45% achieve a good or intermediate outcome — and outcome is modifiable by early treatment and adherence.

The corollary matters as much as the figure: nearly half of patients achieve a good or intermediate outcome. Prognosis should be discussed honestly, not fatalistically — and it is modifiable by early treatment and adherence.

Prognostic factors

Factor	Good prognosis	Poor prognosis
Sex	Female	Male
Age at onset	Older at onset	Young at onset
Onset	Sudden, clearly demarcated	Gradual and insidious
Precipitant	An identifiable stressor	No identifiable stressor
Symptom profile	Prominent POSITIVE symptoms; mood symptoms present	Prominent NEGATIVE symptoms; no mood symptoms
Family history	No family history of schizophrenia	Positive family history
Substance use	No substance misuse	Comorbid substance misuse
Brain imaging	No structural changes	Structural changes present
Premorbid function	Good social and occupational adjustment	Poor premorbid adjustment; long untreated psychosis

Three of these are modifiable — substance misuse, adherence and the duration of untreated psychosis. Those are where clinical effort belongs.

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PART 7

Principles of treatment

Acute, stabilisation and maintenance phases · antipsychotics and their harms · psychosocial treatment · relapse prevention.

The framework of management

At every stage the therapeutic relationship is the foundation. Nothing else works without it.

The four strands

- PHARMACOLOGICAL — antipsychotic medication
- PSYCHOLOGICAL — CBT for psychosis, supportive therapy
- FAMILY — psychoeducation and work on expressed emotion
- SOCIAL — housing, income, occupational rehabilitation, community support
- Delivered together, not in sequence

Psychoeducation — patient and family

- The nature of the illness — that it is a brain illness, not a weakness
- The medication: what it does, how long it takes, what it costs in side effects
- The EARLY SIGNS OF RELAPSE, written down and specific to this patient
- HIGH EXPRESSED EMOTION — criticism, hostility, over-involvement — drives relapse
- What the family can do, and where to get help urgently

High expressed emotion is one of the best-replicated predictors of relapse, and one of the few that family intervention can actually change.

Acute phase — the first goals

Assess and protect

- Make the initial diagnosis: history, MSE, physical examination, investigations
- PREVENT HARM — assess risk to self and others; admit if needed
- Control disturbed behaviour, with de-escalation before medication
- Obtain COLLATERAL information — essential, not optional
- Address precipitating factors, including substances and non-adherence

Treat and engage

- Plan and initiate treatment with the patient wherever possible
- Reduce symptom severity as quickly as is safe
- Aim for rapid return toward optimal functioning
- Build the therapeutic alliance with the patient AND the family
- Begin psychoeducation early — not at discharge

Involuntary admission is sometimes necessary and is governed by the Saudi Mental Health Law; use it as a last resort, and document the reasoning.

First- versus second-generation agents

	First generation (typical)	Second generation (atypical)
Mechanism	Predominantly D2 blockade	D2 blockade plus 5-HT2A antagonism
Examples	haloperidol, chlorpromazine, fluphenazine, trifluoperazine	risperidone, olanzapine, quetiapine, aripiprazole, paliperidone, clozapine
Positive symptoms	Effective	Effective — no clear superiority except clozapine
Negative & cognitive	Little effect; may worsen	Modest benefit at best
Extrapyramidal effects	HIGH — dystonia, akathisia, parkinsonism, tardive dyskinesia	Lower, but dose-dependent and not absent
Metabolic effects	Lower (chlorpromazine is an exception)	HIGH — weight gain, diabetes, dyslipidaemia; worst with olanzapine and clozapine
Prolactin	Raised	Raised with risperidone and paliperidone; aripiprazole spares it

The honest summary: apart from clozapine, the two generations are similarly effective. The choice is made on SIDE-EFFECT profile, patient preference, previous response and cost — not on presumed superiority.

Prescribing principles

Simple regimens, one agent, adequate time, and a decision made with the patient — these four do more for outcome than the choice of molecule.

The rules

- Medication is an ESSENTIAL part of treatment, not the whole of it
- Individualise — response and side effects vary widely between patients
- START LOW, GO SLOW; titrate to the lowest effective dose
- AVOID POLYPHARMACY — one antipsychotic at a time, with rare exceptions
- Simple regimens succeed; complex ones fail on adherence

Timing and choice

- Effect takes TIME: agitation settles in days, psychosis over 2–6 weeks
- Allow an adequate trial — 4–6 weeks at a therapeutic dose — before switching
- Choose on: previous response, family member's response, side-effect profile
- Also on: drug interactions, cost and availability, and PATIENT PREFERENCE
- Involve the patient in the decision — it is the strongest adherence intervention

The commonest prescribing errors are switching too early, adding a second antipsychotic instead of optimising the first, and never asking the patient what they can tolerate.

Harms you must be able to recognise

Whatever specialty you enter, you will meet patients on these drugs. These six are the ones that cause death or permanent harm if missed.

Extrapyramidal effects

Acute dystonia (hours–days) · akathisia (days–weeks) · parkinsonism (weeks–months) · tardive dyskinesia (months–years, often irreversible).

Neuroleptic malignant syndrome

FEVER, lead-pipe RIGIDITY, autonomic instability, altered consciousness, raised creatine kinase. Stop the drug — this is an emergency.

Metabolic syndrome

Weight gain, type 2 diabetes, dyslipidaemia. Worst with olanzapine and clozapine. Monitor weight, BP, HbA1c and lipids from baseline.

QT prolongation

Risk of torsades de pointes — highest with haloperidol (especially IV), ziprasidone and pimozide. Check the ECG and correct K+ and Mg2+.

Hyperprolactinaemia

Amenorrhoea, galactorrhoea, sexual dysfunction, reduced bone density. Common with risperidone; aripiprazole spares prolactin.

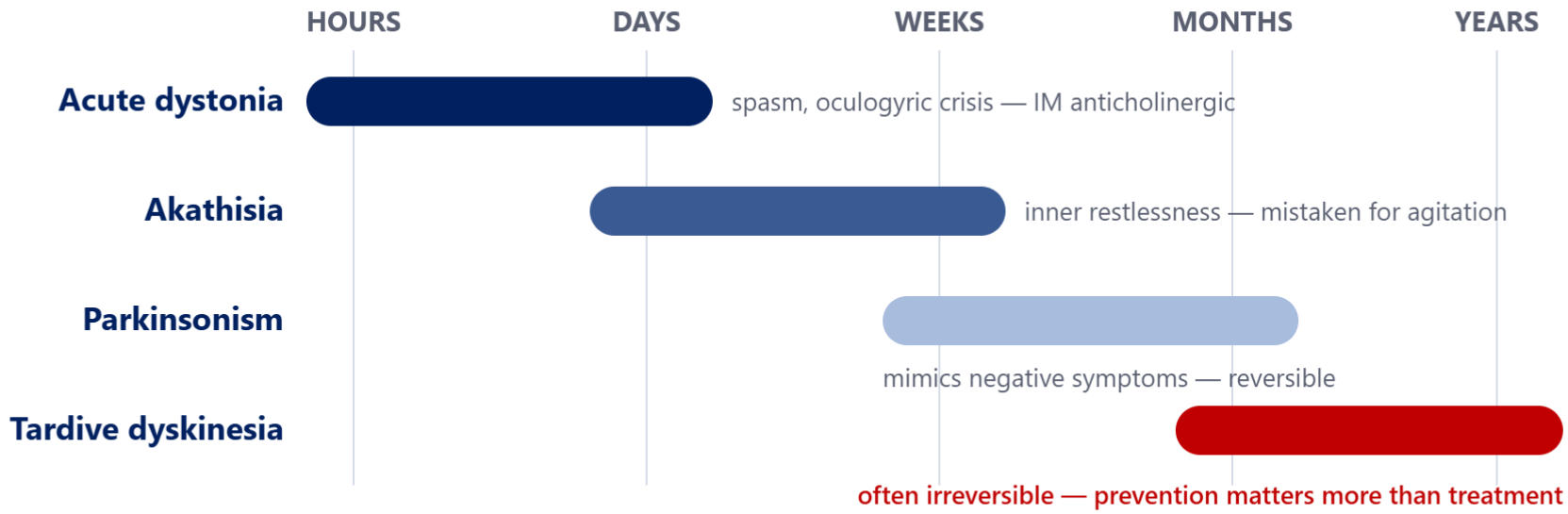
Clozapine-specific

AGRANULOCYTOSIS (mandatory ANC monitoring) · MYOCARDITIS in the first 4–8 weeks · seizures · severe constipation and ileus.

Any patient on an antipsychotic with a FEVER needs an urgent full blood count and an assessment for NMS — two different emergencies with the same first symptom.

Extrapyramidal effects — the early two

Each syndrome has a characteristic time of onset. The timing tells you which one you are looking at.



The timing tells you which syndrome you are looking at.

Akathisia is the most consequential misreading in psychiatry: a restless patient is given MORE antipsychotic, which makes it worse. Ask whether the restlessness is felt inside — that is what distinguishes it from agitation.

Extrapyramidal effects — the later two

The two that develop slowly — and the two most often accepted as part of the illness rather than recognised as drug effects.

Parkinsonism — weeks

- Bradykinesia, rigidity, coarse resting tremor, masked face, shuffling gait
- EASILY MISTAKEN FOR NEGATIVE SYMPTOMS or for depression
- Reduce the dose, or switch to a lower-EPS agent
- Anticholinergics help but add cognitive and constipation burden
- Commoner in older patients and with first-generation agents

Tardive dyskinesia — months to years

- Involuntary choreoathetoid movements — orofacial, tongue, trunk, limbs
- Often IRREVERSIBLE; prevention matters more than treatment
- Risk rises with duration of exposure, age, female sex and first-generation drugs
- Stop or reduce the causative agent where possible
- VMAT2 inhibitors (valbenazine, deutetrabenazine) where available

Drug-induced parkinsonism is the reason to reduce the dose before concluding that a flat, slowed patient has primary negative symptoms — the first is reversible, the second is not.

Clozapine and treatment resistance

The only antipsychotic with proven superior efficacy — and the one with the most demanding safety requirements.

When it is indicated

- TREATMENT-RESISTANT schizophrenia: failure of ≥ 2 adequate antipsychotic trials
- An adequate trial = a therapeutic dose for 4–6 weeks with confirmed adherence
- Also indicated for persistent suicidality in schizophrenia
- And for severe tardive dyskinesia
- Underused and often started years later than it should be

What it demands

- MANDATORY neutrophil monitoring — agranulocytosis is the reason for the register
- MYOCARDITIS in the first 4–8 weeks: fever, chest pain, dyspnoea, raised troponin
- Seizures — dose-related; myoclonus can precede them
- Severe constipation progressing to ILEUS, a recognised cause of death
- Sedation, hypersalivation and marked weight gain

Never restart clozapine after an interruption of more than 48 hours without re-titrating from a low dose — sudden resumption at the previous dose can cause collapse. This is a common error when a patient is admitted to a non-psychiatric ward.

Fever on clozapine

A 31-year-old man has been on clozapine for five weeks for treatment-resistant schizophrenia; his psychosis has improved considerably.

He attends the emergency department with two days of fever to 38.6 °C, malaise and mild chest discomfort. Heart rate 118, no cough, chest clear. The triage note reads: "viral illness — reassure and discharge".

What must be done before he leaves?

A Nothing further — reassure and discharge

B FBC, ECG, troponin, CRP — call psychiatry

C Increase the clozapine dose

D Oral antibiotics and review in a week

Commit to an answer before the next slide.

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B **FBC, ECG, troponin, CRP — call psychiatry**

C Increase the clozapine dose

D Oral antibiotics and review in a week

B — fever on clozapine is never a routine presentation.

Three clozapine emergencies present with fever. AGRANULOCYTOSIS — an urgent full blood count and neutrophil count is mandatory, whatever the story sounds like. MYOCARDITIS — classically weeks 2 to 8, with fever, tachycardia, chest discomfort and raised troponin, CRP and eosinophils; this man is at week 5 with both. NEUROLEPTIC MALIGNANT SYNDROME — rigidity, autonomic instability, altered consciousness, raised CK. Discharging him as viral is the error this slide exists to prevent — and it is diagnostic overshadowing in action: the psychiatric label made the clinician investigate less, not more.

Stabilisation and maintenance

The acute phase is the short one. Most of the illness — and most of the outcome — is decided in these two.

Stabilisation

- Consolidate the response and prevent early relapse
- Help the patient adapt to life in the community
- Provide support and REDUCE STRESS while recovery continues
- Monitor side effects actively — this is when patients quietly stop treatment
- Begin psychosocial intervention; review if the response is incomplete

Maintenance

- Improve level of functioning and quality of life
- Sustain remission and PREVENT RELAPSE
- Continue monitoring side effects and physical health — weight, HbA1c, lipids, ECG
- Reduce modifiable risk: stress from finances, housing, work and relationships
- Address substance misuse and family conflict directly

Physical health monitoring belongs here, not as an afterthought. These patients lose 15–20 years of life predominantly to cardiovascular disease.

Non-pharmacological treatment

Medication reduces positive symptoms. These interventions are what restore function — and they address exactly what the drugs do not.

CBT for psychosis

Works on the distress and conviction attached to residual voices and delusions rather than trying to argue them away. Recommended for persistent symptoms.

Supportive therapy

A consistent, non-confrontational relationship that sustains engagement over years. The foundation of everything else.

Family intervention

Psychoeducation plus work on high expressed emotion. Among the best-evidenced relapse-prevention interventions available.

Group therapy

Social skills training and peer contact — targets the asociality and withdrawal that medication does not touch.

Occupational rehabilitation

Supported employment and structured activity. Work is protective; long-term inactivity deepens negative symptoms.

Social & community support

Housing, income, case management and community follow-up. Practical instability is a leading cause of disengagement and relapse.

Family intervention and supported employment have the strongest functional evidence base of anything on this slide — and both are widely under-provided.

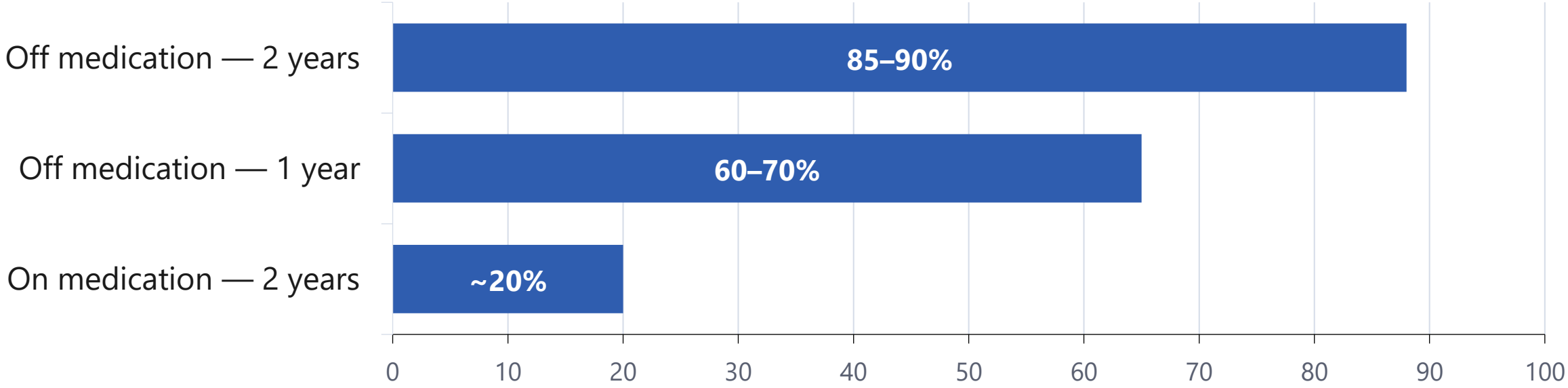
Relapse and why adherence matters

60–70%
relapse within ONE year if antipsychotics are stopped

85–90%
relapse within TWO years if antipsychotics are stopped

~20%
relapse within two years with good adherence

Relapse rates (%) — on treatment versus after stopping antipsychotics



Bars are drawn to the middle of each published range; the labels carry the full range. And adherence itself decays: about 30% of patients stop their medication within one month, and 45% by eight months. Adherence is not a fixed trait — it is something the treatment plan either supports or undermines.

Preventing relapse

The three levers

- ADHERENCE to antipsychotic medication — the single most important factor in preventing relapse and rehospitalisation
- Strong psychosocial support — family, housing, income, follow-up
- ABSTINENCE from substances, particularly cannabis and stimulants
- Add: a written relapse plan with this patient's own early warning signs
- And: continuity — the same team, seen regularly

Why patients stop taking medication

- SIDE EFFECTS — especially weight gain, sedation and sexual dysfunction
- Lack of insight, or feeling well and seeing no further need
- Lack of efficacy, or a complex multi-drug regimen
- COST and availability; distance from the clinic
- Poor rapport with the doctor, and comorbid substance misuse

Consider a LONG-ACTING INJECTABLE early rather than as a last resort — particularly after a relapse caused by non-adherence. Every reason in the right-hand column is something you can ask about and act on.

Board-style questions

#	Question	Answer
1	Psychotic symptoms for 3 weeks with full return to baseline — diagnosis?	Brief psychotic disorder (1 day to < 1 month)
2	Which Criterion A symptoms can NOT satisfy the criterion on their own?	Catatonic behaviour and negative symptoms — one of delusions, hallucinations or disorganised speech is required
3	A first-episode patient reports olfactory hallucinations — next step?	Investigate for an organic cause — classically temporal lobe epilepsy; non-auditory hallucinations are organic until proven otherwise
4	Which dopamine pathway mediates extrapyramidal side effects?	Nigrostriatal — D2 blockade there produces dystonia, akathisia, parkinsonism and tardive dyskinesia
5	Restless, pacing, unable to sit, two weeks after starting risperidone?	Akathisia — reduce the dose or add propranolol; do NOT increase the antipsychotic
6	When is clozapine indicated?	Treatment resistance — failure of ≥ 2 adequate antipsychotic trials; also persistent suicidality and severe tardive dyskinesia

Summary — eight things to remember

On diagnosis

- PSYCHOSIS IS A SYMPTOM, NOT A DIAGNOSIS — always ask why
- Clouded consciousness, fluctuation or visual hallucinations mean DELIRIUM until proven otherwise
- First episode after 40, or any non-auditory hallucination — investigate for an organic cause
- The clock separates the disorders: < 1 month brief · 1–6 months schizophreniform · > 6 months schizophrenia

On illness and treatment

- Criterion A: ≥ 2 symptoms for 1 month, at least one from delusions, hallucinations or disorganised speech
- Mesolimbic D2 blockade treats psychosis; the other three pathways generate the side effects
- Negative and cognitive symptoms determine function — and respond least to drugs
- Adherence is the strongest modifiable predictor of relapse; physical health is the strongest predictor of early death

If only one line survives the year, make it the first: psychosis is a symptom, and the differential begins with the causes you can reverse.

References

Core references

- DSM-5-TR — American Psychiatric Association, 2022
- Kaplan & Sadock's Synopsis of Psychiatry
- Oxford Handbook of Psychiatry
- Stahl's Essential Psychopharmacology — antipsychotic mechanisms
- Maudsley Prescribing Guidelines in Psychiatry

Specific points

- Palmer, Pankratz & Bostwick, 2005 — lifetime suicide risk in schizophrenia (≈ 4.9%)
- CATIE and CUtLASS trials — comparative antipsychotic effectiveness
- NICE CG178 — psychosis and schizophrenia in adults
- Saudi Mental Health Law — involuntary assessment and admission
- Figures given as published RANGES; where sources disagree, both are shown

Where sources disagree, both figures are given — the DSM-5-TR prevalence range alongside the classical 1%, the ≈ 5% suicide estimate alongside the older 10–13%, and the historical subtypes labelled as removed from DSM-5 rather than taught as current.

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Thank You

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Questions welcome. Sources for every figure are on the references slide.